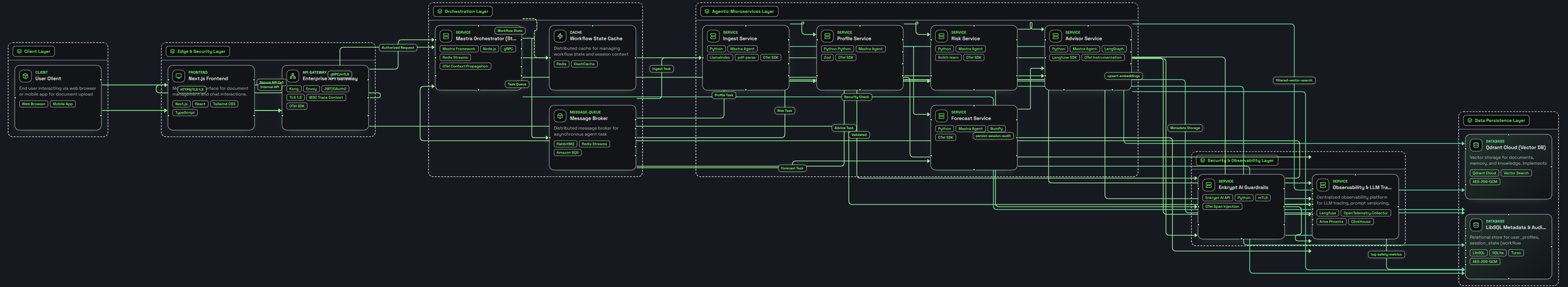

Client Layer
CLIENT
User Client
End user interacting via web browser or mobile app for document upload
Web Browser
Mobile App
Edge & Security Layer
FRONTEND
Next.js Frontend
Mo
HTTPS/TLS 1.3
erface for document management and chat interactions.
Next.js
React
Tailwind CSS
TypeScript
Secure API Call
Internal API
API GATEWAY
gRPC/mTLS
Enterprise API Gateway
Kong
Envoy
JWT/OAuth2
TLS 1.3
W3C Trace Context
OTel SDK
Authorized Request
Orchestration Layer
SERVICE
Mastra Orchestrator (St...
Workflow State
Mastra Framework
Node.js
gRPC
Redis Streams
OTel Context Propagation
Task Queue
CACHE
Workflow State Cache
Distributed cache for managing workflow state and session context
Redis
ElastiCache
MESSAGE-QUEUE
Message Broker
Distributed message broker for asynchronous agent task
RabbitMQ
Redis Streams
Amazon SQS
Ingest Task
Agentic Microservices Layer
SERVICE
Ingest Service
Python
Mastra Agent
LlamaIndex
pdf-parse
OTel SDK
Profile Task
Risk Task
SERVICE
Profile Service
Python Python
Mastra Agent
Zod
OTel SDK
Security Check
Advice Task
Validated
Forecast Task
SERVICE
Risk Service
Python
Mastra Agent
Scikit-learn
OTel SDK
SERVICE
Forecast Service
Python
Mastra Agent
NumPy
OTel SDK
persist-session-audit
SERVICE
Advisor Service
Python
Mastra Agent
LangGraph
Langfuse SDK
OTel Instrumentation
upsert-embeddings
filtered-vector-search
Metadata Storage
Security & Observability Layer
SERVICE
Enkrypt AI Guardrails
Enkrypt AI API
Python
mTLS
OTel Span Injection
SERVICE
Observability & LLM Tra...
Centralized observability platform for LLM tracing, prompt versioning,
Langfuse
OpenTelemetry Collector
Arize Phoenix
ClickHouse
log-safety-metrics
Data Persistence Layer
DATABASE
Qdrant Cloud (Vector DB)
Vector storage for documents, memory, and knowledge. Implements
Qdrant Cloud
Vector Search
AES-256-GCM
DATABASE
LibSQL Metadata & Audi...
Relational store for user_profiles, session_state (workflow
LibSQL
SQLite
Turso
AES-256-GCM